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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 6: Impacts of Transportation Systems-I

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Dept. of Civil Engineering



Contents

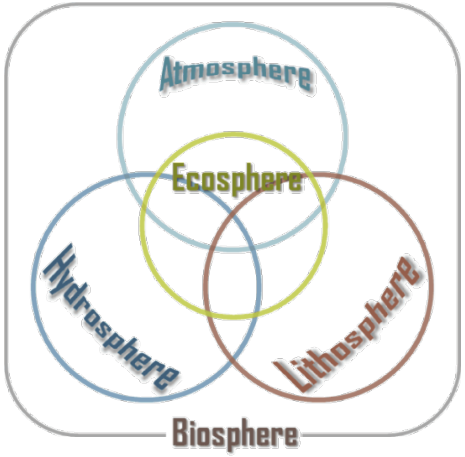
- Transport-Environmental Link
- Energy Consumption and Emissions by Transport Sector
- Environmental Impacts of Transport
- Transport Sector & Climate Change



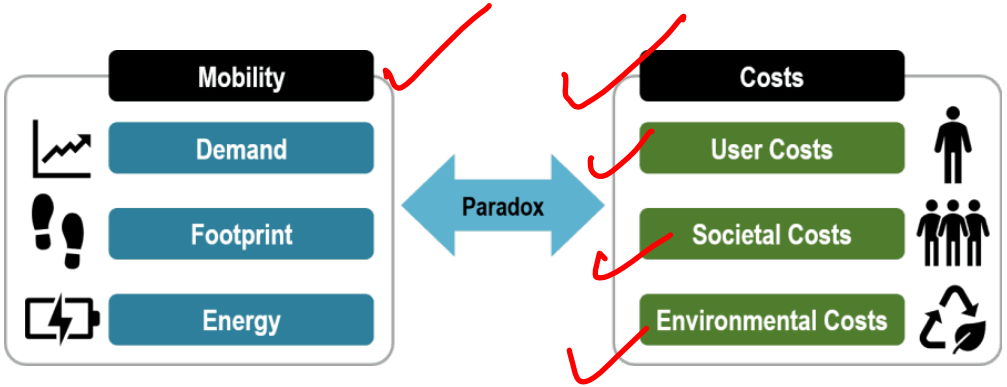
The Transport-Environmental Link

- Issue of transportation and the environment is paradoxical in nature.

- Transportation conveys substantial socio economic benefits, but at the same time transportation is impacting environmental systems.



The Environmental System



The paradox of mobility and its costs

Source: (Jean-Paul Rodrigue, 2020)

The Transport-Environmental Link

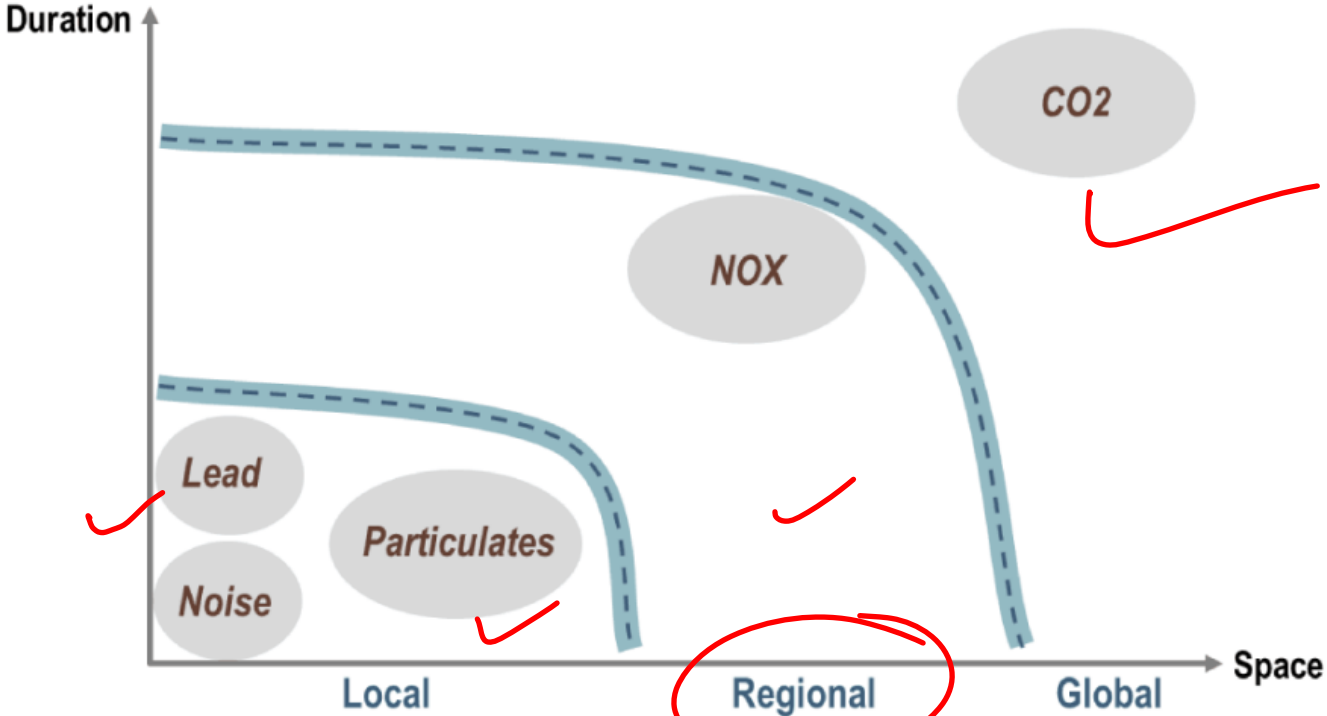
- The environmental dimensions of Transportation are related to the causes, activities, outputs and the results of the Transport systems.

- Level of contribution:
  - Transport activities contribute among other anthropogenic and natural causes, directly, indirectly, and cumulatively to environmental problems.

- Scale of Impact:
  - Transport activities contribute at different geographical scales to environmental problems, ranging from local (noise and CO emissions) to global (climate change), even continental/ national/regional problems (smog and acid rain)

Source: (Jean-Paul Rodrigue, 2020)

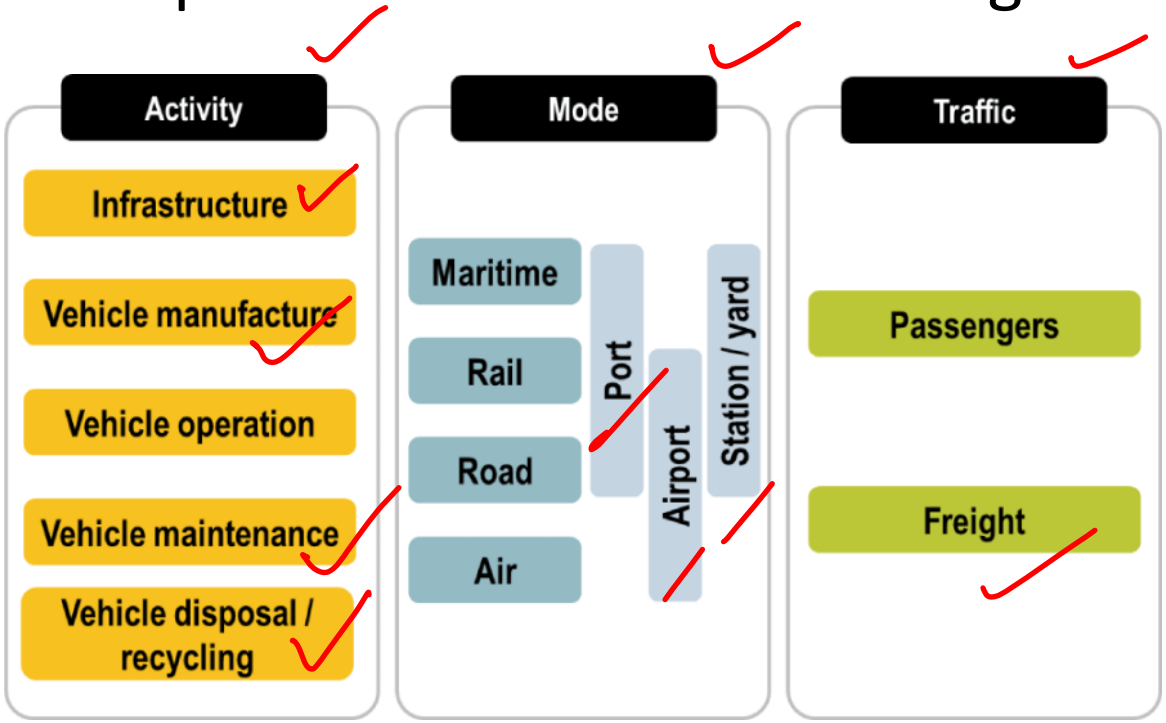
Transportation activities affecting the Environment



Spatial and Durational Environmental effects of pollutants

Source: (Jean-Paul Rodrigue, 2020)

Transportation activities affecting the Environment

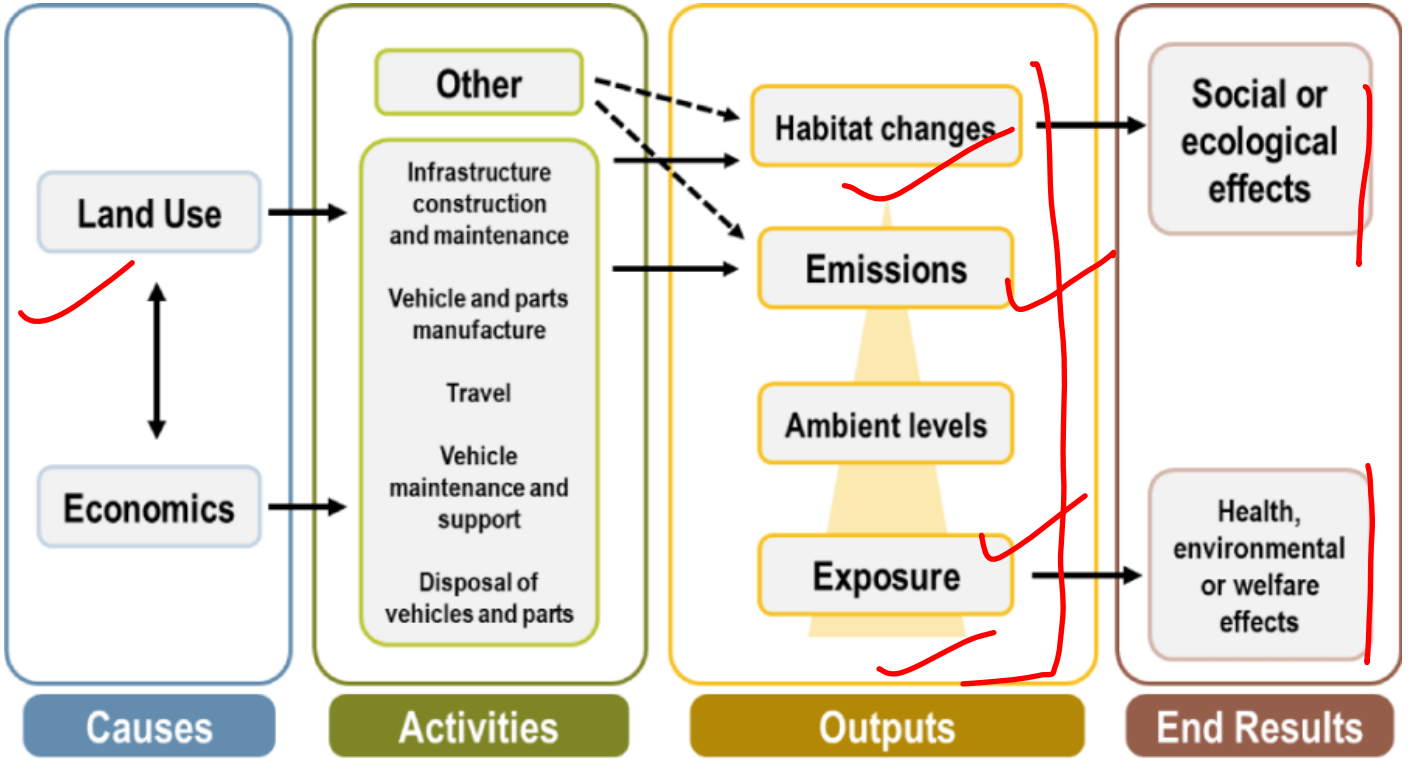


Transportation activities affecting the Environment

- Infrastructure: Resources, energy and land.
- Vehicle manufacture: Resources and energy consumed
- Vehicle operation: Fossil fuel consumption
- Vehicle maintenance: Consumption and disposal of vehicle parts, lubricants etc.
- Vehicle disposal: Disposal after use

Source: (Jean-Paul Rodrigue, 2020)

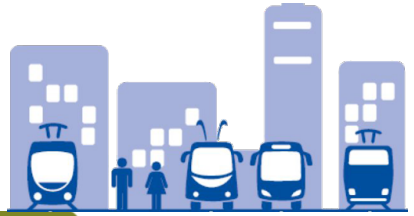
Environmental Dimensions of Transportation



Source: (Jean-Paul Rodrigue, 2020)

Transport and Energy Consumption

- Transportation activities are significant energy consumers, providing mobility to passengers and freight.
- Accounts for about 20-25% of world energy use.
- Strong correlation with the level of development
- 87% of World's total power production is derived from Fossil fuels

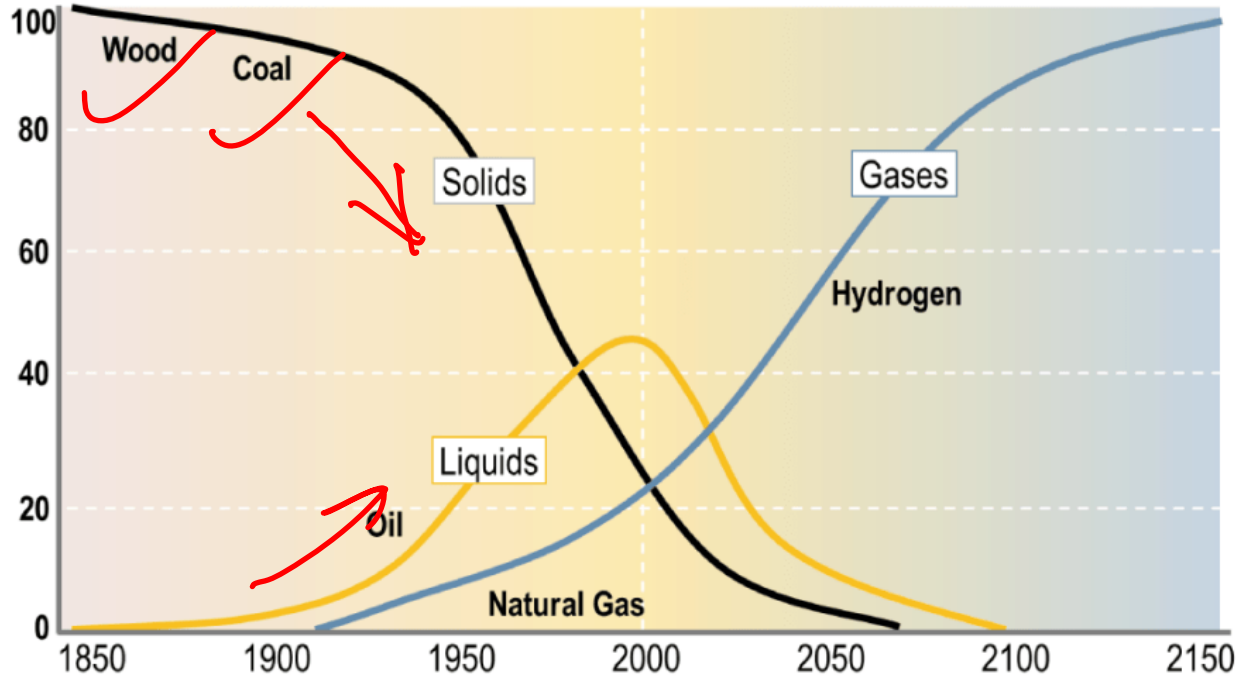


Non-Renewable	ENERGY	Renewable
Chemical • Fossil fuels (Combustion) **Nuclear** • Uranium (Fission of atoms)	**Chemical** **Nuclear** **Gravitational** **Electromagnetic** **Mechanical** **Thermal** **Electrical**	**Chemical** • Muscular (Oxidization) **Nuclear** • Geothermal (Thermal conversion) • Fusion (Fusion of hydrogen) **Gravitational** • Tidal (Mechanical) **Indirect Solar** • Biomass (Photosynthesis) • Wind and hydraulic (Mechanical) **Direct Solar** • Photovoltaic cell (Conversion) • Solar thermal (Thermal conversion)

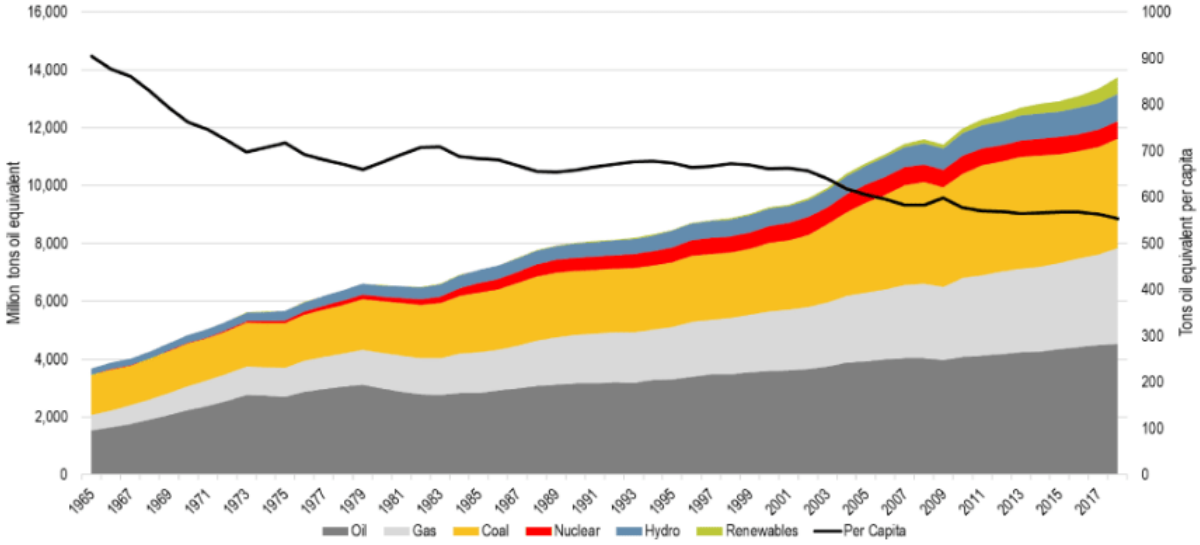
Source: (Jean-Paul Rodrigue, (2020)

Global Energy Systems Transition (% of market)

Source: (Jean-Paul Rodrigue, 2020)



World Energy Consumption

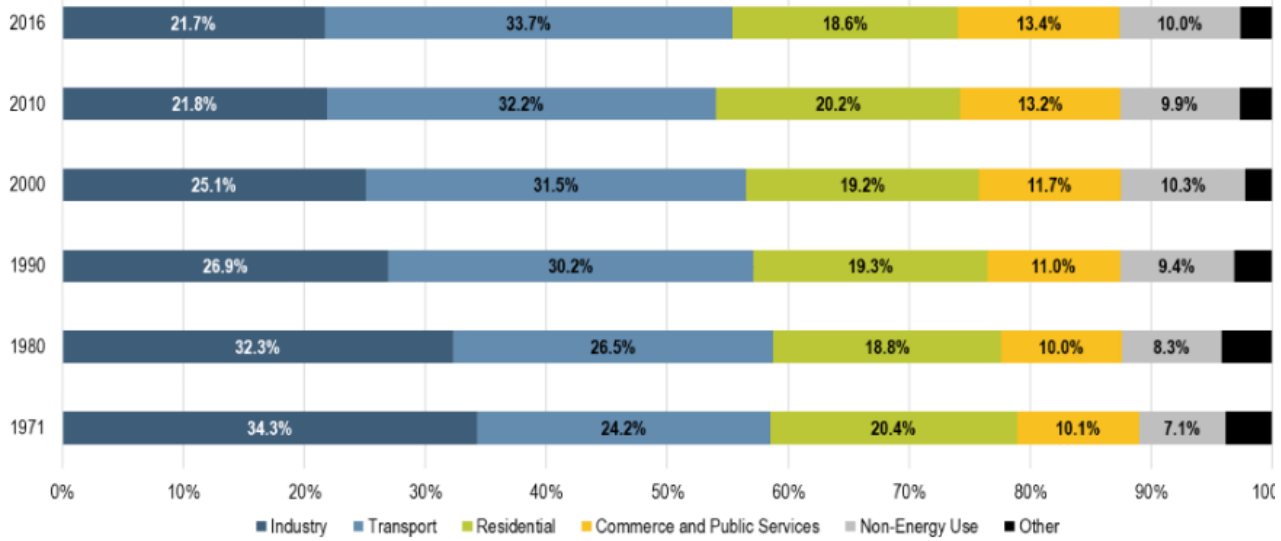


World Energy Consumption (1965-2018)

- Coal consumption declined from 39% (1965) to 28% (2016).
- Oil consumption declined from 40% (1965) to 33% (2016).
- Natural gas consumption increased from 15% (1965) to 24% (2016).
- Consumption of Renewables increased to 3% (2016).

Source: (Jean-Paul Rodrigue, 2020)

Energy Consumption by Sector

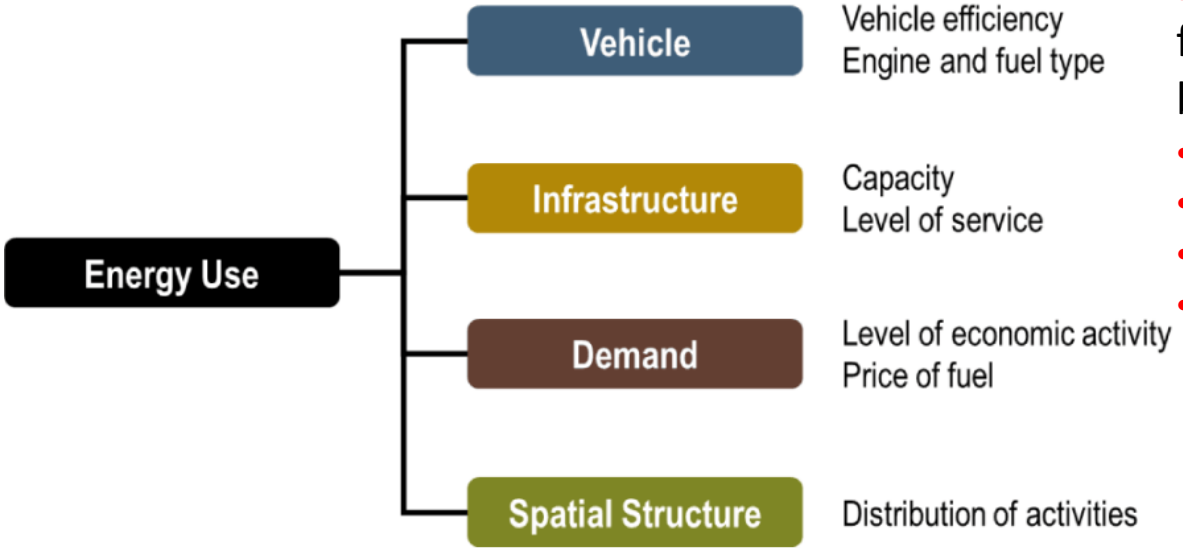


Energy Consumption by Sector, OCED Countries

- The Transport sector consumed 24.2% of the energy in 1971, and climbed to 33.7% in 2016.

Source: (Jean-Paul Rodrigue, 2020)

Factors of Energy Use by Transportation



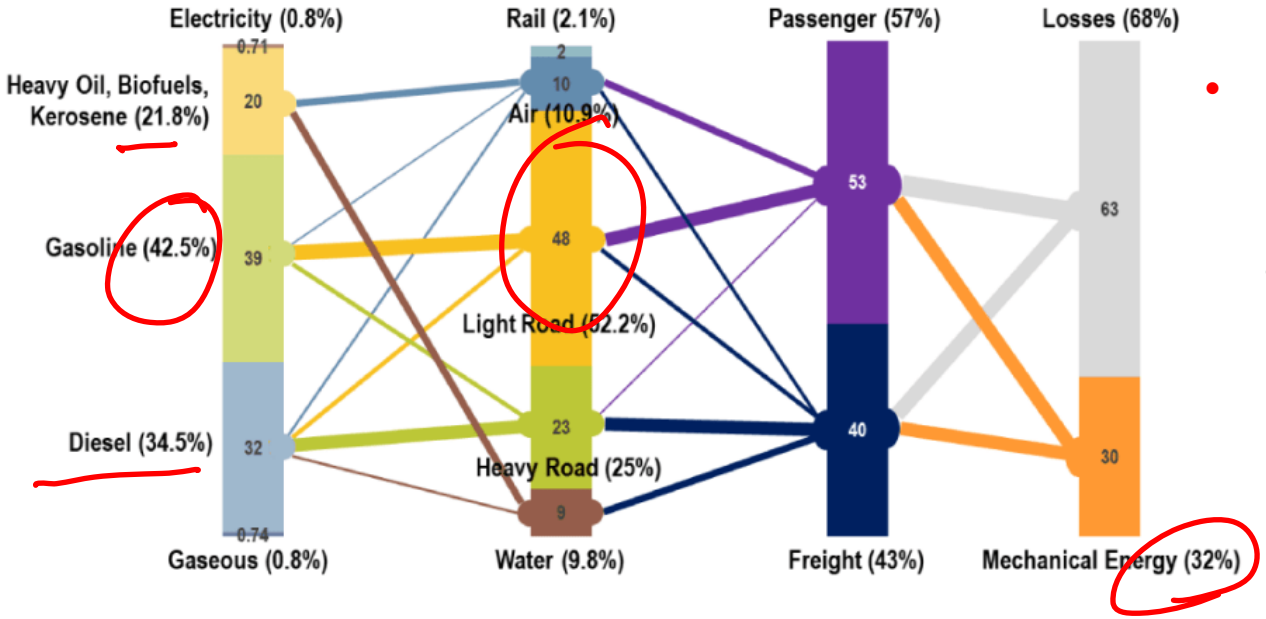
Four major factors jointly influence fuel consumption and energy use by transportation

- Vehicles (Fuel efficiency)
- Infrastructure (Transport efficiency)
- Demand (Economic activity)
- Spatial Infrastructure (Average transport distance)

Energy Consumption by Sector, OCED Countries

Source: (Jean-Paul Rodrigue, 2020)

Energy Consumption by Fuel Type in Transport Sector

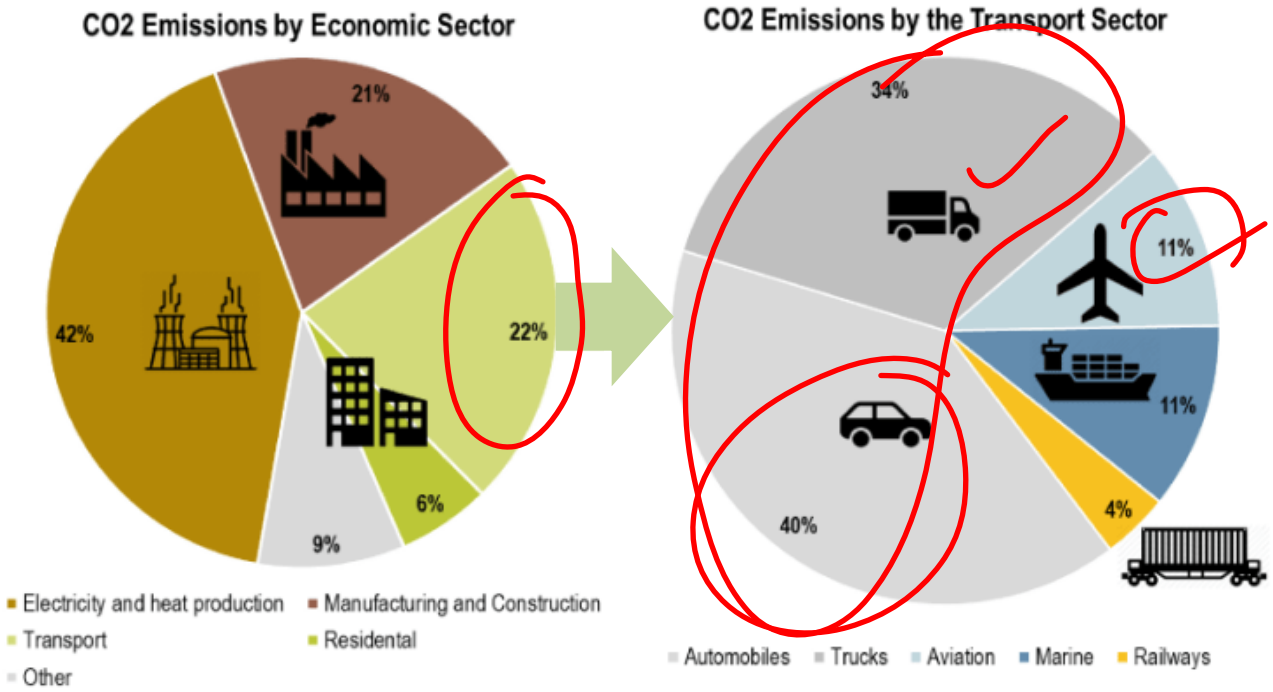


Energy Consumption by Fuel Type

- Oil-related fuels accounted for 94% of the energy used by transportation modes.
- Most used fuel is gasoline (43.5%), followed by diesel (34.5%) heavy oil, biofuels, and kerosene (21.8%). Electricity and gases (e.g. LNG).

Source: (Jean-Paul Rodrigue, 2020)

Global GHG emission by Transport Sector



Global GHG emission by the Transport Sector

- Energy generation is the most significant contributor of GHG emissions globally.
- Transportation sector accounts for 22% of the Global GHG emissions.

Source: (Jean-Paul Rodrigue, 2020)

Average CO_2 Emissions by Transport modes

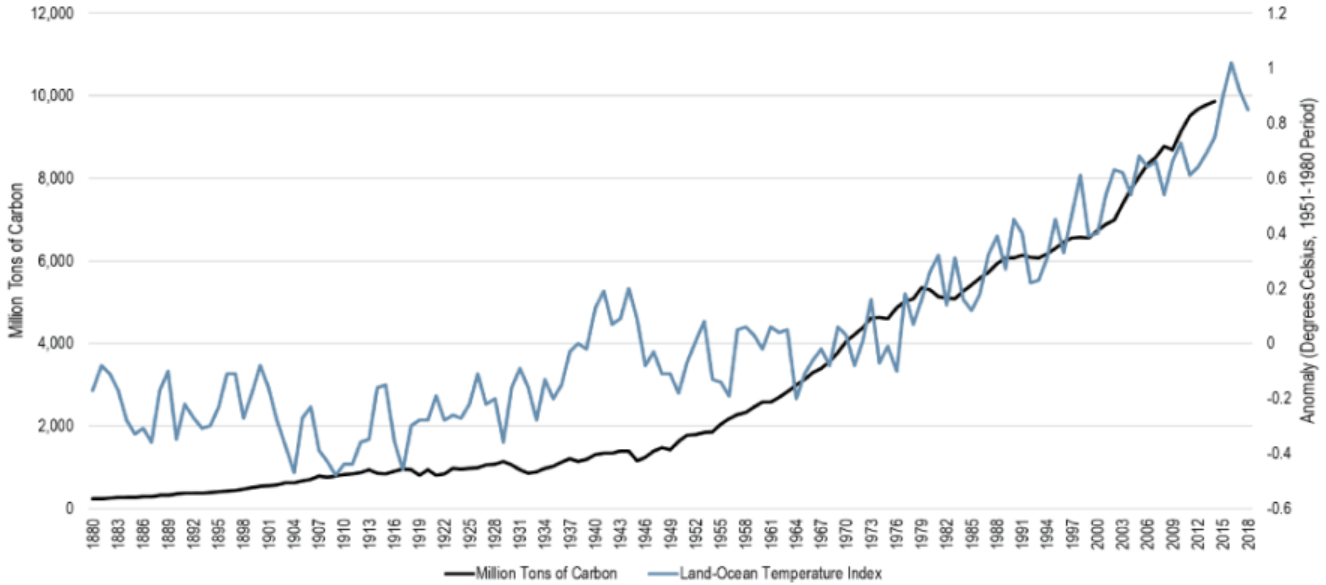




Average CO_2 Emissions by Passenger and Freight Transport modes

Source: (Jean-Paul Rodrigue, 2020)

Carbon Emissions from Fossil fuel Burning & Average Global Temperature

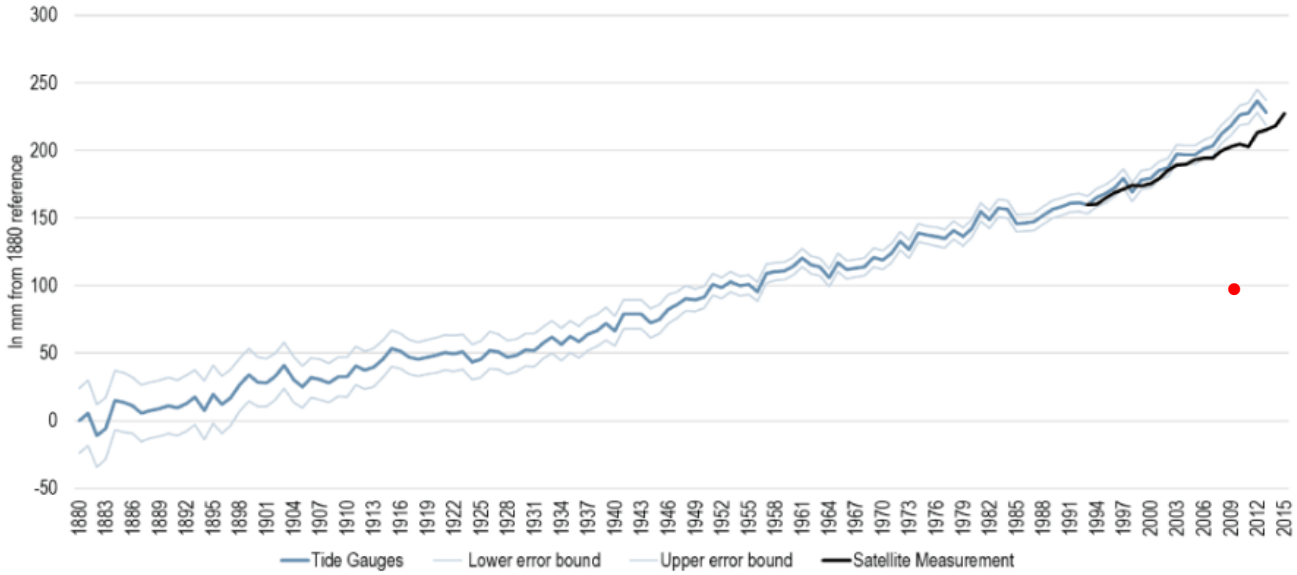


Average Global Temperature and World Carbon Emissions From Fossil Fuel Burning, (1880-2018)

Source: (Jean-Paul Rodrigue, 2020)

- Sharp increase in CO_2 levels after 1960s.
- China and India have become the world's largest Carbon emitters in the early 21st century.

Global Mean Sea Level Change



Global Mean Sea Level Change (1880-2015)

- The main factors that contributed to the rise in global mean sea level are
  - increase in the quantity of water in the oceans
  - increase in the global average temperatures.
- Increased risk of flooding and damage to coastal infrastructures.

Source: (Jean-Paul Rodrigue, 2020)

Environmental Externalities of Transportation

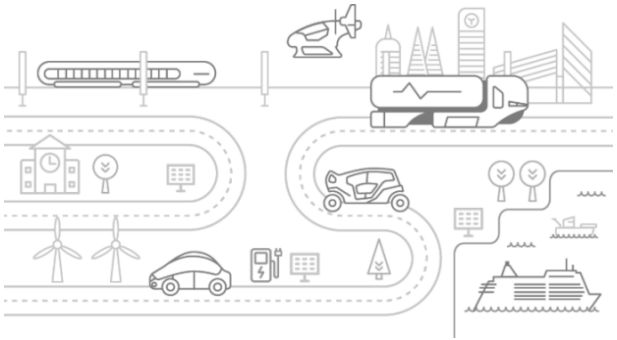
Supra and infrastructure	Vehicle emissions; CO2 (30%), CO (70-90%), SO2 (5%), NOx (45-50%), HC/VOC (40-50%), Particulates (25%), O3 (indirect), Smog (indirect), Acid rain (10-30%), CFCs (20%), Lead (30%)
Labor Productivity	
Agricultural Productivity	
Commercial fishing	Air pollution fallouts, Marine vessels discharges and spills, De-icing of infrastructure, Runoffs, Construction and maintenance of infrastructure, Dredging (80%)
Recreational facilities	
Water purification	
Accidents/Spills	Marine vessels spills, Accidental and intentional releases (Hazmat); Road (84%), Rail (12%), Air (3%) and Maritime (1%)
Property values	Vehicle emissions; Road (70%), Rail (10%) and Air (20%)
Public health	Vehicle emissions, During transport
Damage to ecosystems	Vehicle emissions, during transport, Infrastructure

Environmental Externalities of Transportation

Source: (Jean-Paul Rodrigue, 2020)

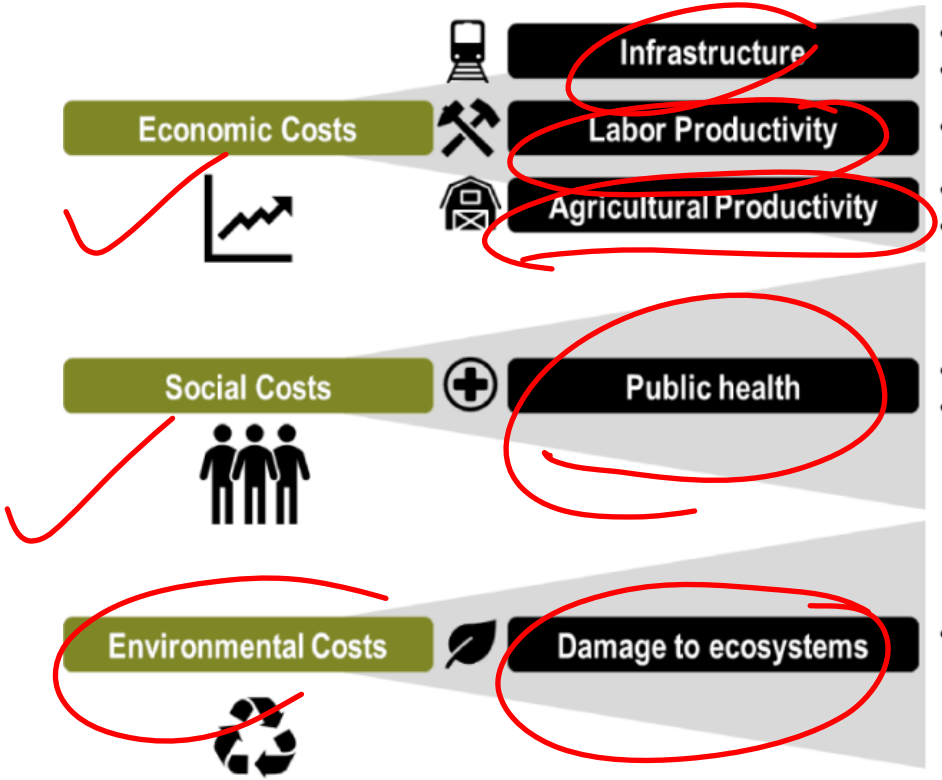
Environmental Externalities: Emissions, AQ & Climate Change

- Transportation activities support increasing mobility demands for passengers and freight, notably in urban areas.
- But transport activities have resulted in growing levels of motorization and congestion.
- As a result, the transportation sector is becoming increasingly linked to environmental problems, mainly linked to Climate change and Air quality.



Source: (Jean-Paul Rodrigue, 2020)

Externalities of Air Pollution



- Loss of useful life (amortization)
- Replacement and restoration
- People-hours-wage losses
- Yield decrease
- Biomass restoration time
- Medical services costs
- Burden of diseases
- Biological diversity and sustainability

Source: (Jean-Paul Rodrigue, 2020)

Impacts on Air Quality & Human Health





- Highway vehicles, marine engines, locomotives, and aircraft are the sources of pollution in the form of gas and particulate matter emissions.

- They affect air quality and cause damage to human health.

- The most common include lead (Pb), carbon monoxide (CO), nitrogen oxides (NOx), silicon tetrafluoride (SF6), benzene and volatile components (BTX), heavy metals (zinc, chrome, copper, and cadmium) and particulate matters (ash, dust).

Source: (Jean-Paul Rodrigue, 2020)

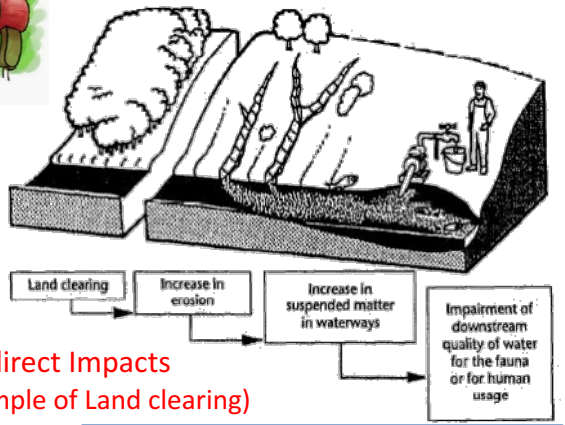
Health & Environmental Impacts of Transport

- Direct Impacts
  - Noise pollution
  - Emission of Carbon Monoxide



Direct Impacts

- Indirect Impacts (Secondary or Tertiary)
  - Higher consequence than direct impacts
    - Particulates from Incomplete combustion
    - Respiratory and cardiovascular problems



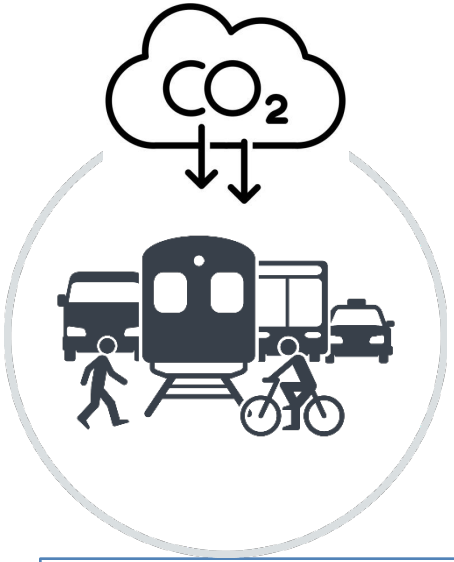
Indirect Impacts
(The example of Land clearing)

- Cumulative Impacts
  - Additive, multiplicative or synergic consequences
    - Climate change

Source: (Jean-Paul Rodrigue, 2020)

How Transportation impacts Climate ?

- By use of vehicles powered by fossil fuels such as coal and oil such as cars, buses, two and three wheelers, steam-powered trains, boats etc.
- As technology and vehicles increased the CO_2 generated by vehicles increased.
- The CO_2 concentration released over decades have caused a greenhouse effect, warming the Earth at rapid pace and thus leading to Climate change.
- The Average temperature of Earth has increased by 0.8^0C since the early 20th century.



Source: (Sarin Abraham et. al., 2012)

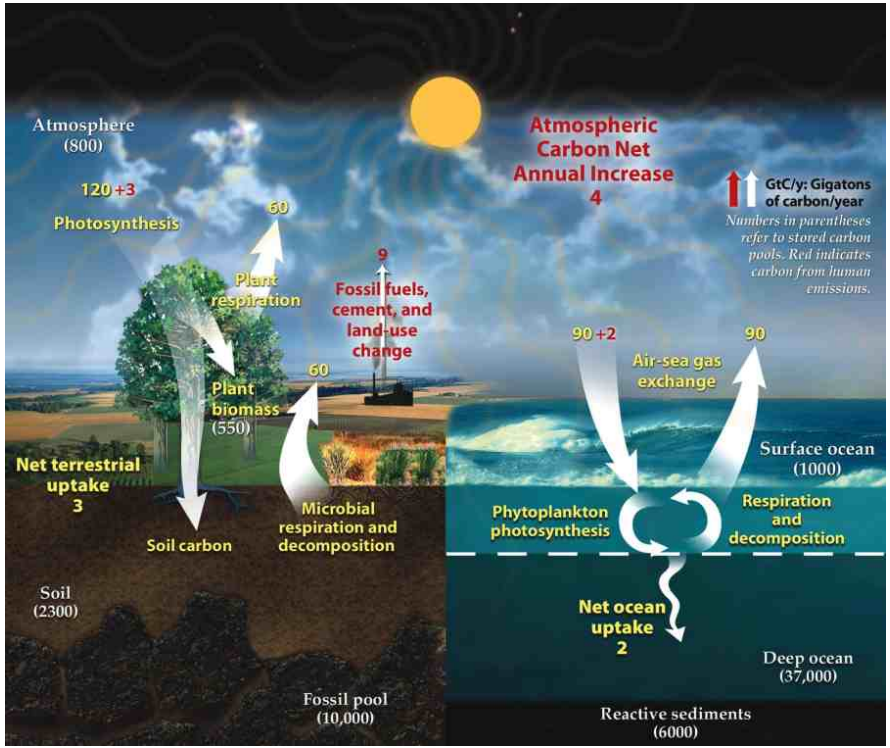
Impacts of Climate change on Transportation

	Operations	Infrastructures
 Heat waves	• Impacts of lift-off load limits on shorter runways. • Limits on periods of construction activity.	• Thermal expansion of bridges. • Pavement integrity and softening. • Deformation of rail tracks.
 Rising sea levels	• Frequent interruptions of coastal low lying road, rail and air traffic due to storm surges.	• More frequent flooding of infrastructure (and potential damage) in low lying areas. • Erosion of infrastructure support. • Changes in harbor facilities to accommodate higher tides and surges.
 Intensity of precipitation	• Increase in weather related delays and disruptions, particularly road and air transport.	
 More frequent hurricanes	• Frequent interruptions of air services. • Frequent and extensive evacuations of coastal areas. • Debris of road and rail infrastructures.	• Greater probability of infrastructure failure. • Greater damage to port infrastructures.
 Increase in arctic temperatures	• Longer shipping season. • More ice-free ports in northern regions. • Availability of trans-arctic shipping routes.	• Damage to infrastructure because of the thawing of the permafrost. • Shorter season for ice-roads.

- Elements associated with climate change have potential impacts on transport operations and infrastructures

Source: (Jean-Paul Rodrigue, 2020)

Transportation and Carbon Cycle



Transportation and Carbon Cycle

- The figure illustrates the potential contribution of human activities, including transportation, to different geographical scales of the carbon cycle.
- Numbers in parenthesis refer to stored Carbon pools (in Gigatons of Carbon/year (GtC/y)
- Red indicates Carbon from Human emissions

Source: (Jean-Paul Rodrigue, 2020)

Scale of Carbon emissions from Transport modes

 Car 1 Person: 175 grams per km per passenger	 Aeroplane: 101 grams per km per passenger	 Motorbike: 119 grams per km per passenger
 Car 2 Person: 88 grams per km per passenger	 Bus: 89 grams per km per passenger	 Moped: 85 grams per km per passenger
 Car 3 Person: 59 grams per km per passenger	 Tram: 65 grams per km per passenger	 Bike/Walk: 0 grams per km per passenger
 Car 4 Person: 44 grams per km per passenger	 Train: 54 grams per km per passenger	

Carbon emissions by various Modes of Transport (Emissions per Person)

- Walking and Cycling are the best modes of transport
- Sharing a vehicle reduces emissions compared to single trips
- Mass/ Public Transport have less emissions/trip

Source: (https://blogmech.com/transportation-and-climate-change-carbon-emissions-by-transport-type/)

Environmental Kuznets curve (EKC)

- According to EKC hypothesis, in the early stages of economic growth, pollution and degradation of the environment increase.
- Beyond some level of income per capita, the trend reverses, so that at high income levels economic growth leads to environmental improvement.

Source: David I. Stern, 2004.

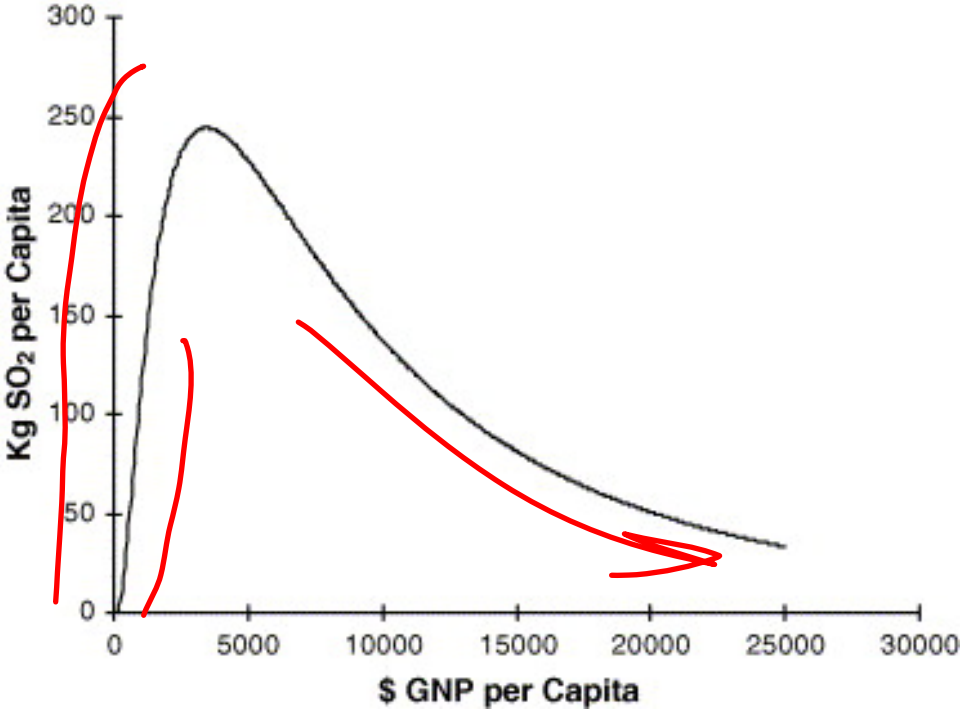


Fig. Environmental Kuznets curve for sulfur emissions.
Source: Panayotou (1993) and Stern, Common, and Barbier (1996).

Conclusions

- Transportation activities have significant impacts on the Environment.
- The emissions from Transport sector is a major contributor to Global warming and Climate change.
- The GHG emissions from transport sector can be reduced by measures such as:
  - reduced use of private automobiles
  - increased use of public transport facilities
  - use of alternate clean fuels for transport
  - promotion of Non-motorized modes
  - integrated Transport development.



References

- David I. Stern (2004). The Rise and Fall of the Environmental Kuznets Curve. World Development 32 (8), 1419-1439. https://doi.org/10.1016/j.worlddev.2004.03.004

- D.I Stern, M.S Common, E.B Barbier (1996). Economic growth and environmental degradation: The environmental Kuznets curve and sustainable development. World Development, 24, pp. 1151-1160

- Jean Paul Rodrigue (2020). "The Geography of Transport Systems" Fifth Edition, New York: Routledge, 456 pages. ISBN 978-0-367-36463-2, https://transportgeography.org/geography-of-transport-systems-5th-edition/.

- Kadiyali, L. R. (2017). "*Traffic engineering and transport planning*", Khanna Publishers.

- Panayotou, T. (1993). Empirical tests and policy analysis of environmental degradation at different stages of economic development. Working Paper WP238, Technology and Employment Programme, International Labour Office, Geneva.

- Sarin Abraham, Ganesh, K., Senthil Kimar, A, and Yves Ducq (2012). "*Impact on Climate Change due to Transportation Sector- Research Prospective",* International Conference on Modeling Optimisation and Computing (ICMOC-2012), Procedia Engineering, Vol. 38, pp. 3869-3879.

Thank You